

Reading: Linux Commands

Estimated duration:

Due to its robust nature, Linux is a powerful operating system often preferred by developers and administrators. Linux allows users to configure the operating system tailored to their needs and work patterns. It is also free to use because it is open source.

Learning objectives:

- Explain the purpose of common Linux commands
- Use a terminal window to input Linux commands

At the heart of accessing and utilizing the Linux operating system is the terminal window where users type their commands rather than using a mouse and the graphical user interface, or GUI for short, like that of macOS or Windows. Note, though, that macOS and Windows also have command line interfaces, but most of their users typically interact with the OS using the GUI.

Linux commands introduction

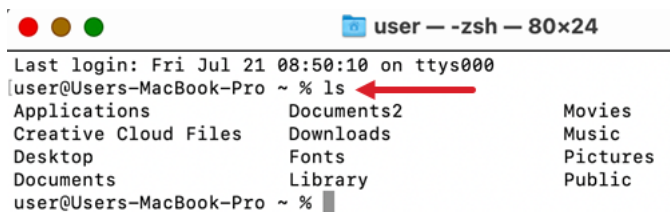
The most often used Linux commands can be divided into categories. To execute a command in Linux, you type a command in the terminal window, usually two to six letters. Common types of functionalities used in an operating system include:

- navigate through and operate on files and directories
- create and edit text files
- view file content
- administration, such as permissions and configuring a network
- install software and updates, and
- compress and archive directories and files

Linux has commands for all of these. You can find a list of commands commonly used by administrators in the cheat sheet accompanying this module.

The "list" and "change directory" commands

One of the most common uses of an OS is file and directory navigation. In Windows or macOS, this is equivalent to clicking on folders to navigate a folder structure and identify files or apps you want to work with. One of the most common things you will want to do is see the contents of a folder. You use the `ls` command for this. So, if you are in your documents folder and type `ls` the output is a list of files and folders, as shown here.



```

user — zsh — 80x24
Last login: Fri Jul 21 08:50:10 on ttys000
user@Users-MacBook-Pro ~ % ls
Applications      Documents2        Movies
Creative Cloud Files  Downloads        Music
Desktop           Fonts            Pictures
Documents         Library          Public
user@Users-MacBook-Pro ~ %

```

Another common command is the `cd` command which stands for "change directory." To use the `cd` command, type

```
cd \<path/directory name\>
```

So, for example, if you type

```
cd Applications
```

where "Applications" is a directory in your working directory, the window will change directories to the Applications folder.

Other commonly used commands

File and directory operations commands

- **ls** - List Files and Directories: With the `ls` command, users can obtain a detailed listing of files and directories in their current location. This command reveals valuable information such as file permissions, ownership, size, and modification dates.
- **pwd** - Print Current Working Directory: The `pwd` command displays the full path of the current working directory, making it easy to ascertain your location within the file system.
- **mv** - Move and Rename: When you need to reorganize your files or give them a new name, the `mv` command comes to your rescue. It facilitates both file moving and renaming operations.
- **cp** - Copy Files and Directories: The `cp` command is indispensable for duplicating files or creating backups. It allows you to copy files and directories from one location to another.
- **rm** - Remove Files and Directories: The `rm` command lets you delete files and directories permanently. Caution is advised, as there is no undo functionality.
- **chmod** - Change File Permissions: The `chmod` command is used to control who can read, write, or execute your files. It adjusts file permissions to enhance security.
- **chown** - Change Ownership: To transfer ownership when another user group needs the file the `chown` command is employed.

System management and administration commands

- **su** or **sudo** - Switch User and Super User: Elevate your privileges with `su` (Switch User) or `sudo` (Super User Do) commands. They empower you to execute commands with heightened permissions, which is crucial for system-wide changes.
- **apt-get** (**sudo**) - Package Management (Debian-based): `apt-get` is your gateway to managing software packages on Debian-based systems like Ubuntu. It assists in installing, upgrading, and removing software with ease.
- **yum** vs **apt** - Package Management Dilemma: A choice between `yum` (RPM-based) and `apt` (Debian-based) package managers often arises. Each caters to different Linux distributions, such as CentOS and Ubuntu.

Networking and connectivity commands

- **ip** - Network Interface and Routing: The `ip` command grants insight into network interfaces and routing, enabling the configuration and troubleshooting of networking settings.
- **df** - Display Disk Space: The `df` command provides a comprehensive view of disk usage across various file systems to keep tabs on available disk space.
- **grep** - Search with Regular Expressions: When searching for specific patterns within files, the `grep` command, armed with regular expressions, proves invaluable.
- **find** - Search for Files and Directories: The `find` command lets you traverse directory hierarchies in search of files and directories matching specific criteria.
- **dig** - Query DNS Servers: Need to resolve domain names or fetch DNS records? The `dig` command interfaces with DNS servers, providing valuable information for network management.

Text manipulation and editing commands

- **cat** - Concatenate and Display Content: Display the contents of a file using the `cat` command, essential for quick peeks into file details.
- **nano** - Text Editing in the Terminal: When a lightweight text editor is needed right from the terminal, `nano` makes quick edits a breeze.

Data archiving and transfer commands

- **tar** - Archive and Compress: The `tar` command combines files into an archive and optionally compresses them, facilitating efficient storage and transportation.
- **rsync** - Synchronize Files and Directories: For seamless file synchronization between local and remote systems, the `rsync` command proves its worth.

Tab key, paths, and directories

You do not need to type the full name of the directory. The terminal window can do that for you by typing the first part of its name and then pressing the Tab key. So, the command:

```
cd App<tab>
```

Would result in the command line displaying:

```
cd Applications
```

Then, press Enter, and you change directories to the Applications folder.

If you want to access a file in a parent directory, type:

```
cd ../<folder name>
```

The tab key works the same by autocompleting the folder name.

Command history

To access previously typed commands, repeatedly press the up arrow on your keyboard to find the command you want to use. Using the up arrow on the command line serves to access previous commands starting with the most recent command entered.

Summary

This text introduces Linux as a powerful, flexible, and open-source operating system widely used by developers and system administrators. It emphasizes the importance of the terminal and command-line interface for interacting with Linux, contrasting it with GUI-based systems like Windows and macOS. The document outlines learning objectives and categorizes commonly used Linux commands, covering file and directory navigation, system administration, networking, text editing, and data archiving. Key commands such as `ls`, `cd`, `chmod`, `sudo`, `grep`, and `tar` are explained, along with productivity features like tab completion and command history. Overall, the content provides a foundational overview of essential Linux commands and terminal usage.



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